

Contract-Centered Agentic Runtimes

From responsibility boundaries to executable 5W1H data-use contracts

**Capabilities evolve. Capacity grows.
Data identity, authority and evidence must remain checkable.**

Status: executable mechanism and synthetic checks; real-task and scaling studies remain open

Yaxiao Liu · Pengbo Liu · Yiwon Liu · Yihua Guan · Zhenghe Hou · Jiaxing Song
PwC China AI Center / Tsinghua University

| Task success does not establish system manageability

Enterprise work crosses business, model, infrastructure and data-governance boundaries

Business

Is the result useful?
Are all required fields correct?
How much human rework remains?

Runtime

Does capacity change behavior?
Can failures be recovered?
Are enforcement costs acceptable?

Data

Is the source version correct?
Is access still authorized now?
Can claims be traced to evidence?

Business evidence and runtime evidence must be reported separately.

| v29 makes the data boundary executable

01 Responsibility contracts

Skill / Harness / Scaffold
Independent external
data governance

Design

02 Data-use boundary

5W1H+Which predicates
Revalidation and evidence
binding

**Executable, bounded reference
model**

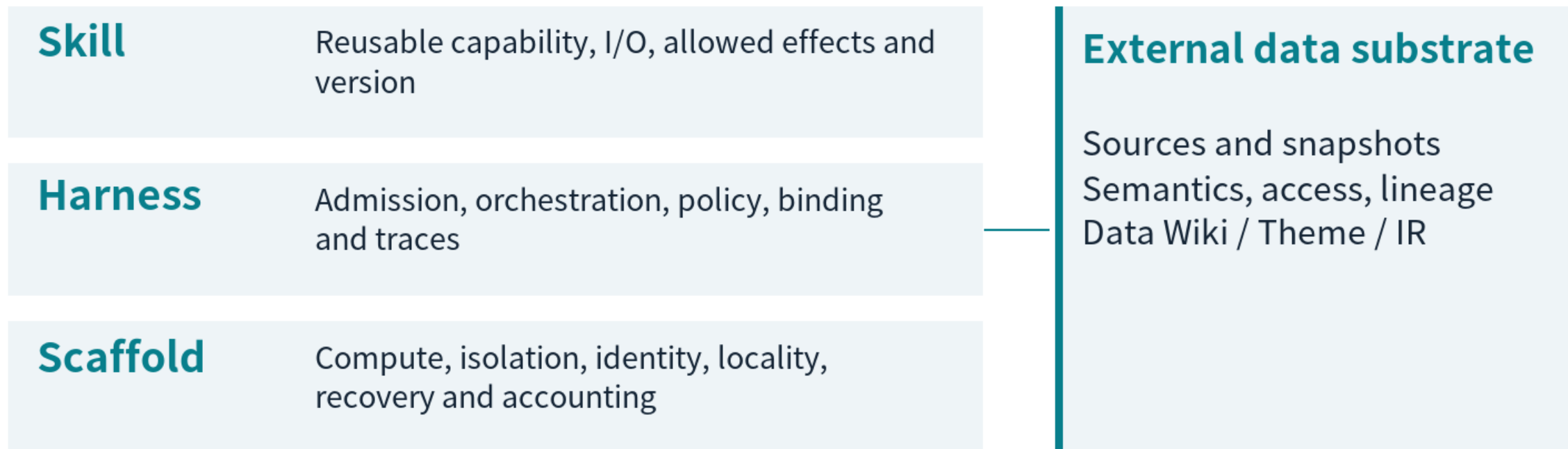
03 Falsifiable evaluation

Capability × capacity
experiment
Quality, violations and
total cost

Confirmatory study not yet run

Four responsibility objects; three runtime layers

The data substrate sits outside the stack with independent semantic ownership



One team may implement several objects; their contracts remain distinct.

P1: can capability and capacity be separated?

Test interaction effects, semantic stability and enforcement cost

Same workload, model, policy and source snapshot

$c0 \times s0$
R / Q / E

$c0 \times s1$
R / Q / E

$c1 \times s0$
R / Q / E

$c1 \times s1$
R / Q / E

Measure every cell

R: latency, throughput, timeouts

Q: every-requirement satisfaction

E: admission, checks and evidence cost

P1 remains untested in a real runtime experiment in this study.

A minimal counterexample: same columns, different meaning

Illustrative scenario; not measured data

Source A: group revenue

Fields: product_id, revenue

Scope: consolidated / USD / 2026 Q2

Source B: subsidiary revenue

Fields: product_id, revenue

Scope: local entity / EUR / 2026 Q1

Relevant retrieval and executable joins do not establish admissibility.

Data Wiki, Theme Wiki and IR answer different questions

Data Wiki

What is this data?
Source, version, fields,
policy
Read and extraction
operators

IR

How may this task use it?
5W1H+Which
Evidence, compatibility,
fallback

Theme Wiki

What must the output
satisfy?
Fields, units, periods
Quality criteria and verifier

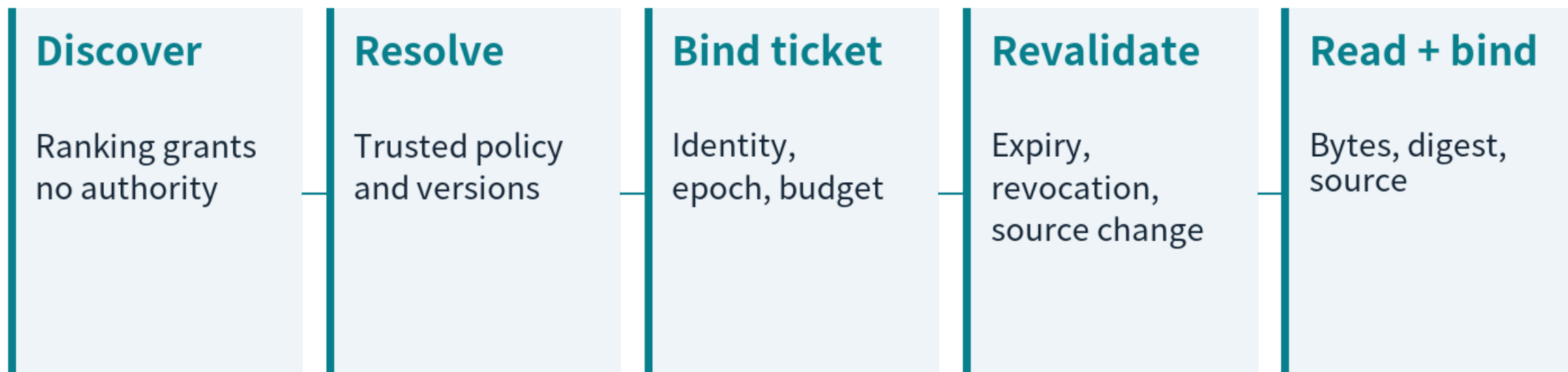
Descriptions are metadata; sources and trusted policy remain authoritative.

| Turn 5W1H+Which into checkable conditions

When	Business period, expiry, current time	Expiry while queued
Where	Access domain and delivery boundary	Internal data released externally
Who	Trusted identity and current access	Revocation after admission
What	Units, entity scope, business meaning	Same name, different meaning
Why	Approved integration rationale	Invented integration rationale
How	Allowed operations and budget	Read request becomes write
Which	Relations, versions, keys, lineage	Wrong relationship or snapshot

A populated field does not mean its constraint is enforced.

From candidates to execution tickets to evidence



Typed failures: STALE / UNAUTHORIZED / SOURCE_CHANGED / REVALIDATE

| Revocation during queuing: is admission enough?

t0 Admission

t1 Revocation

t2 Execution



Policy epoch = 7
Ticket created



Policy epoch = 8
Old ticket invalidated



Revalidate
Reject or re-admit

The model uses a lock; distributed adapters still need a consistency protocol.

Change impact follows explicit dependencies

Source / policy change

IR invalidated

Artifact revalidation

Condition: complete dependencies

Hidden views, shared prompts and mutable aliases can break locality

Current model: conservative

Global epoch; no measured fine-grained maintenance savings

| An existing file is not a verified deliverable

Theme completion predicate

- All required fields present
- Artifact digest matches this run
- Evidence dependencies remain valid
- Specified verifier has passed

Keep states distinct

- Completed and verified
- Partial
- Rejected
- Expired; revalidation required

A content hash proves byte identity, not truth or semantic entailment.

Executed evidence: finite contract conformance

One revenue fixture × ten binary fault dimensions; exhaustive within this domain

1,024

Declared cases

1,023

Expected rejections

0

Oracle disagreements

Five lifecycle checks pass: revocation, replacement, expiry, replay and revalidation

This is finite conformance, not retrieval-quality or scalability evidence.

Evidence boundary: what can we say today?

Supported now

- Executable data-boundary specification
- Ten-dimensional fault domain checked
- Five lifecycle checks passed
- Code, cases and hashes are inspectable

Not established

- Better real-task accuracy
- Lower enterprise lifecycle cost
- Distributed safety or scalability
- P1 support or conference readiness

Recent work: execution, research and delivery

Prime Agent

Persistent REPL, recursive sessions, tree-wide accounting
Transfer: control budget and state
Limit: scores do not establish P1

AutoResearch

Independent review and claim-evidence binding
Transfer: require real artifacts
Limit: examples are not universal reliability

Apodex 1.1

Evidence synthesis and controlled delivery
Transfer: manifests and validity
Limit: its scaling axes differ from P1

Recent work: indexing, routing and source access

VoiceMem

Separate upper index and backend

Transfer: isolate index and routing

Limit: retrieval latency is not end-to-end latency

NeoHorse-1

Predicted demand $\neq$ served route

Transfer: freeze models; log rerouting

Limit: one update is not sustained self-improvement

Beyond Top-K

Source structure, units and time matter

Transfer: strong BM25 + source controls

Limit: small, single-document study

| Methodological repairs in v29

Insufficient measurement Unresolved coverage, calibration or intervals → inconclusive

Clear obligation violation Adequate evidence and violation beyond bound → conditional-engineering

All obligations pass Then classify P1 using semantic, interaction and cost intervals

Margins and power Decision-derived margins; p95 power needs tail-aware simulation

Next experiment: isolate contract enforcement

Central causal contrast

A: source-preserving prose catalog
B: same catalog + enforcement
Fix candidates, tools, model, budget
Hold out sources, templates and time

Report together

All fields correct with valid evidence
Unauthorized / stale release
False refusal on valid tasks
Build + maintenance + online cost

Add strong retrieval baselines and ablations; separate navigation from governance.

Internal pilot: start with one bounded workflow

Proposed plan; no internal pilot is claimed

Phase 1: specify

Select output fields and sources

Define units, periods and access

Label valid and refusal cases

Phase 2: integrate

Integrate one real read adapter

Implement execution-time checks

Audit mediation with independent logs

Phase 3: evaluate

Run fixed-budget comparisons

Report refusal, quality and total cost

Expand only after evidence gates pass

Venue choice follows the contribution and evidence

Dates checked on 2026-09-10; the current manuscript still needs empirical studies

SIGMOD / PACMMOD

Data contracts, heterogeneous data, maintenance cost

Abstract Oct 10; paper Oct 17, AoE

OSDI 2027

Implemented runtime, monitoring, scale and failures

Abstract Dec 1; paper Dec 8, 22:59 UTC

ICSE / FSE

Contract evolution, change impact, maintainability

ICSE 2027 closed; FSE dates unverified

Choose the primary research question, then fund the evidence it requires.

| Three decisions for the internal discussion

- 01 Choose the question** Data-use contracts first, or a full capability–capacity study?
- 02 Choose the evidence setting** Which real workflow has clear sources, authority and verifiable outputs?
- 03 Define acceptance criteria** Who owns acceptable quality degradation, false refusal and total cost?

v29 turns an architectural claim into something executable and falsifiable.

| Reading and reproduction

Five new readings plus one recheck; links point to the original papers

Prime Agent <https://arxiv.org/html/2608.23552v1>

AutoResearch <https://arxiv.org/html/2608.17906v1>

Apodex 1.1 <https://arxiv.org/html/2608.23283v1>

VoiceMem <https://arxiv.org/html/2608.26005v1>

NeoHorse-1 <https://arxiv.org/html/2609.08183v1>

Beyond Top-K <https://arxiv.org/html/2608.06305v1>

Reproduce: artifact_v29/evaluate.py + test_contract.py
Raw evidence: fault-cases.csv / summary.json; P1 not evaluated